

DIHWorld Audio Technical Reference

Source-verified family and parameter reference

Current authority: 2026-08-01

Customer delivery formats: macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64. Standalone applications or other source build targets are not customer deliverables unless a specific installer explicitly lists them.

VOCAL MIC SOURCE Technical Manual

Vocal, microphone, and source-tone shaping

Build or protect the source before downstream color, dynamics, and spatial processing.

Included products: GOLD, SILVER, WPR, Aegis Nova, TRIPLER, EchoLocate

Routing authority: Lead vocal: GOLD -> AEGIS NOVA -> SILVER. Background vocal: GOLD -> TRIPLER -> EchoLocate.

Canon baseline

Normalized control 1 equals 0.809016994 (PHI/2). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

GOLD

Master-series microphone voicing that combines source-array blend with silver air, copper body, iron weight, gold presence, platinum authority, and nickel focus.

Declared product: GOLD | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool
sourcearray	Source / Mic Array	Factory
AG_AIR	Ag Air	Factory
CU_BODY	Cu Body	Factory
FE_GLUE	Fe Weight	Factory
AU_DRIVE	Au Presence	Factory
PT_AUTHORITY	Pt Authority	Factory
NI_FOCUS	Ni Focus	Factory

Artwork authority

assets/gold_bg.png

SILVER

Precision microphone voicing for truth, focus, grounding, air, and transient definition with protected input, mix, and output control.

Declared product: SILVER | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool
AG_TRUTH	Truth Field	Factory
NI_FOCUS	Precision	Factory
CU_GROUND	Ground	Factory
AIR	Air	Factory
TRANSIENT	Transient	Factory
OUTPUT	Output	Factory
INPUT_GAIN	Input	Factory
MIX_BLEND	Mix	Factory

Artwork authority

assets/silver_bg.png

WPR

Phi ribbon microphone channel with transformer family, core material, winding, lamination, polar pattern, ribbon mass, proximity, and magnetic behavior.

Declared product: WPR | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
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Parameter surface

ID	Display name	Type
DRIVE	Transformer	Float
ARCHITECTURE	Ribbon Tone	Float
DEPTH	Ribbon Mass	Float
WIDTH	Side Width	Float
FOCUS	Polar Focus	Float
ENERGY	Output Trim	Float
CLARITY	Air Edge	Float
MIX	Ribbon Blend	Float
CONVERGENCE	Proximity	Float
IRON_DRIVE	Iron Drive	Float
FLUX	Magnetic Flux	Float
HEADROOM	Core Headroom	Float
PRI_BIAS	Primary Bias	Float
SEC_COLOR	Sec Coloring	Float
LOAD	Winding Load	Float
PERMEABILITY	Permeability	Float
HYSTERESIS	Hysteresis	Float
GEOMETRY	Iron Geometry	Float
gainmatch	Gain Match	Bool
LAMINATION	Lamination Type	Choice
TRANSFORMER_FAMILY	Transformer Family	Choice
CORE_MATERIAL	Core Material	Choice
WINDING_TOPOLOGY	Winding Topology	Choice
CORE_AGE	Core Age	Choice
POLAR_MODE	Polar Pattern	Choice
OUTPUT	Output	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

source/WPR.png

Aegis Nova

Vocal shield for plosive and sibilance control with a restrained drive stage and parallel mix.

Declared product: AEGIS_NOVA | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
gainmatch	Gain Match	Bool
power	Power	Bool
plosive	Plosive	Float
sibilance	Sibilance	Float
drive	Drive	Float
mix	Mix	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

assets/aegisNova.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT	Incoming peak activity	Confirms source level reaching the shield.
NOVA	Detected plosive/sibilant activity	More light means stronger shield intervention.

TRIPLER

Three-voice vocal harmonizer with key, scale, chord, pitch correction, voice gain, pan, fine tune, and Phi field control. The current left/right voices use the repaired high-quality overlap-add pitch path for clean stereo harmonies.

Declared product: TRIPLER | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call not detected; integration review required before release claims
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
DIATONIC	DIATONIC	Referenced
CHORD	CHORD	Referenced
KEY	KEY	Referenced
SCALE	SCALE	Referenced
L_GAIN	L_GAIN	Referenced
R_GAIN	R_GAIN	Referenced
C_GAIN	C_GAIN	Referenced
L_PAN	L_PAN	Referenced
R_PAN	R_PAN	Referenced
BYPASS	BYPASS	Referenced
PITCH_CORRECT	PITCH_CORRECT	Referenced
PC_AMOUNT	PC_AMOUNT	Referenced
FINE_TUNE	FINE_TUNE	Referenced
PHI_FIELD	PHI_FIELD	Referenced
L_ON	L_ON	Referenced
R_ON	R_ON	Referenced
OUT_GAIN	OUT_GAIN	Referenced
GAINMATCH	GAINMATCH	Referenced

Artwork authority

assets/TRIPLER.png

The repaired left/right pitch path uses OSAMP 8, overlap-add normalization, and fractional spectral-bin interpolation to remove the prior light distortion while preserving the three-voice design.

EchoLocate

Three-position source-placement processor for spread, drift, microphone gain, proximity, pan, and reflected-space mix.

Declared product: EchoLocate | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
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Parameter surface

ID	Display name	Type
MODE	MODE	Referenced
SPREAD	SPREAD	Referenced
DRIFT	DRIFT	Referenced
MIX	MIX	Referenced
MIC1_GAIN	MIC1_GAIN	Referenced
MIC2_GAIN	MIC2_GAIN	Referenced
MIC3_GAIN	MIC3_GAIN	Referenced
MIC2_PROX	MIC2_PROX	Referenced
MIC3_PAN	MIC3_PAN	Referenced
OUT_GAIN	OUT_GAIN	Referenced
channelmode	channelmode	Referenced
dualmonolink	dualmonolink	Referenced

Artwork authority

resources/echolocate.png

DYNAMICS GUARD Technical Manual

Envelope, transient, pressure, and containment control

Control entry, impact, sustain, and pressure without hiding the source.

Included products: ANUBIS, OTIS, DEBO, LOOKOUT

Routing authority: Use on individual tracks or buses according to Smart Position. DEBO may be used for parallel impact.

Canon baseline

Normalized control 1 equals 0.809016994 ($\text{PHI}/2$). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

ANUBIS

Dynamics gate and guard with threshold, ratio, attack, hold, release, range, a named stereo host Sidechain bus, independent Sidechain Key and Sidechain Duck operation, protected I/O, and parallel mix.

Declared product: ANUBIS | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call not detected; integration review required before release claims
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
input	Input	Float
mix	Mix	Float
output	Output	Float
power	Power	Bool
threshold	Threshold	Float
ratio	Ratio	Float
attack	Attack	Float
hold	Hold	Float
release	Release	Float
range	Range	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
extkey	Sidechain Key	Bool
scduck	Sidechain Duck	Bool
gainmatch	Gain Match	Bool

Artwork authority

Assets/Images/Anubis.png

Meter and visualizer semantics

Display	What it measures	How to read it
IN	Main-input RMS energy	Shows the program feeding the gate.
GATE	Current gate passage gain	OPEN is full passage; lower values mean attenuation.
KEY	External sidechain RMS energy	Moves only when a host sidechain is routed.
DUCK	Sidechain-driven gain reduction	More fill means deeper ducking of the main signal.
OUT	Post-gate RMS energy	Compare with IN to judge actual attenuation.
Transfer graph	Threshold, envelope, and Range floor	Shows where the signal opens and how far closure may fall.
Correlation waves	Left/right phase relationship	Gold is coherent, purple is spread, red warns of anti-phase risk.

ANUBIS exposes a named optional stereo host Sidechain bus. Sidechain Key routes that bus to the detector/envelope while the main input remains the only audible program path; Sidechain Duck independently applies the detected envelope as ducking.

OTIS

3-6-9 macro dynamics processor: Micro Peak controls short events, Meso Density shapes body, and Macro Level governs the long envelope.

Declared product: OTIS | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
trumpet_l	Micro Peak	Float
trumpet_c	Meso Density	Float
trumpet_r	Macro Level	Float
makeup_gain	Makeup Gain	Float
gainmatch	Gain Match	Bool

Artwork authority

source/Otis.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT	Input peak in dB	Confirms level feeding all three time scales.
RESPONSE	Strongest active 3/6/9 reduction	Shows the greatest current reduction across Micro, Meso, or Macro.
OUTPUT	Output peak in dB	Use with INPUT and Gain Match for fair comparison.
3 / Micro activity	Fast transient reduction	Responds to short peaks and front edges.
6 / Meso activity	Body and phrase-density reduction	Responds to medium-duration energy.
9 / Macro activity	Long-envelope reduction	Responds to sustained level movement.

Factory programs

OTIS PRESET

DEBO

Pressure-activated dynamics processor with nine updated canonical programs using the restored iCloud names. The approved powered-on default is unchanged; the named bank uses a warmer depth/pressure progression without runtime preset clamping, topping out at 2.2 dB Maximize.

Declared product: DeBo | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** not detected | **Dual-mono link:** not detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call not detected; ceiling review required
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
POWER	Power	Bool
FLOWER	Flower of Life	Bool
GAINMATCH	Gain Match	Bool
INPUT	Input	Factory
DRIVE	Drive	Factory
MAXIMIZE	Maximize	Factory
DEPTH	Depth	Factory
CLARITY	Clarity	Factory
OUTPUT	Output	Factory

Artwork authority

Assets/Images/Debo_state1_emerge.png

Meter and visualizer semantics

Display	What it measures	How to read it
RMS	Sustained signal body	Shows average energy driving the pressure model.
GUARD PRESSURE	Combined density/containment demand	Higher values mean DEBO is working harder.
PEAK	Fast peak relative to ceiling	Approaching the top warns of limited remaining space.
GOVERNOR	Gain held back by protection	More activity means stronger safety containment.
Pressure window / face	RMS-driven pressure stage	The visual advances with real pressure; red is the ceiling warning state.

The current nine-program bank restores the canonical iCloud names without runtime clamping. VOCAL HOLD is value-identical to the preferred older preset; RED LINE WATCH is the strongest factory stage at 4.2 dB Maximize. The final safety curve uses raTanhPhi behavior rather than hard preset flattening.

Factory programs

NO BRICKS [Stereo]; VOCAL HOLD [Stereo]; BRICK TAP [Stereo]; WALL PRESSURE [Stereo]; SHOULDER CHECK [Stereo]; NO SCISSORS [Stereo]; BREAKTHROUGH [Stereo]; RED LINE WATCH [Stereo]; PHI GUARD [Stereo]

LOOKOUT

Dual-envelope transient/sustain shaper. Attack and Sustain directly move the level of the transient and sustain portions of the signal, up to +/-18 dB each, ahead of the DIH tone-color stage. Its Eye of Heru state also gives live advisory guidance on transient/sustain balance while it shapes.

Declared product: LookOut | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
power	Power	Factory
attack	Attack	Factory
sustain	Sustain	Factory
gainmatch	Gain Match	Factory

Artwork authority

assets/lookout_sentinel_bg.png

Meter and visualizer semantics

Display	What it measures	How to read it
Transient	Detected front-edge energy	Use to judge whether Attack is emphasizing or restraining hits.
Sustain	Detected body and tail energy	Use to judge Sustain changes to ring, room, and body.
Alert	Potential transient/sustain imbalance	Higher values suggest the balance needs attention.
Miss	Underrepresented event energy	Indicates material that may be disappearing.
Stress	Over-concentrated envelope pressure	Reduce shaping intensity when stress remains high.
Composure	Stable transient/sustain balance	Higher is calmer and more controlled.
Eye of Heru	Combined advisory state	It reports balance; it does not move controls automatically.

Factory programs

NEUTRAL SENTINEL [Stereo]; VOCAL FRONT GUARD; DRUM SNAP LOCK; KICK BODY ANCHOR; PERCUSSION EDGE; ROOM TAIL WATCH; MIX BUS COMPOSURE; FINAL PEAK SENTINEL

BASS LOW END Technical Manual

Bass, sub, foundation, and low-end transformation

Build weight and depth while retaining definition, placement, and controlled output.

Included products: MARIANA, MONOLITH, Depth Boost Pro, NTRPL

Routing authority: Bass route: MARIANA -> NTRPL -> MONOLITH. Depth Boost Pro is an optional focused stage.

Canon baseline

Normalized control 1 equals 0.809016994 (PHI/2). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

MARIANA

Sub and foundation processor with depth, sub pressure, kick definition, target mode, parallel mix, and an optional kick-key sidechain that ducks only MARIANA's added low-end layer.

Declared product: MARIANA | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Low End Source | **Channel mode:** not detected | **Dual-mono link:** not detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
power	Power	Bool
depth	Depth	Float
pressure	Sub Pressure	Float
foundation	Foundation	Float
kick_define	Kick Define	Float
mode	Target Mode	Choice
input	Input	Float
mix	Mix	Float
gainmatch	Gain Match	Bool
sidechain_enable	Sidechain Key	Bool
sidechain_duck	Sidechain Duck	Float
sidechain_recovery	Key Recovery	Float

Artwork authority

assets/Mariana.png

Meter and visualizer semantics

Display	What it measures	How to read it
DEPTH	Current trench-depth control state	Higher values extend the added low-end field deeper.
PRESSURE	Current sub-pressure control state	Higher values add more low-frequency pressure.
FOUNDATION	Current sustained-foundation control state	Higher values add more continuous low-end support.
IMPACT	Current kick-definition control state	Higher values preserve more front-edge definition.
LOW ENERGY	Detected low-frequency program energy	Shows the bass energy entering MARIANA's field.
BEDROCK	Detected sustained foundation energy	Shows the slower body underneath transient bass events.
KEY	Detected 35-160 Hz sidechain energy	Moves only when KEY is enabled and a host sidechain is routed.
DUCK GR	Reduction of MARIANA's added low-end layer	More activity means the external key is clearing more generated pressure; dry bass remains audible.

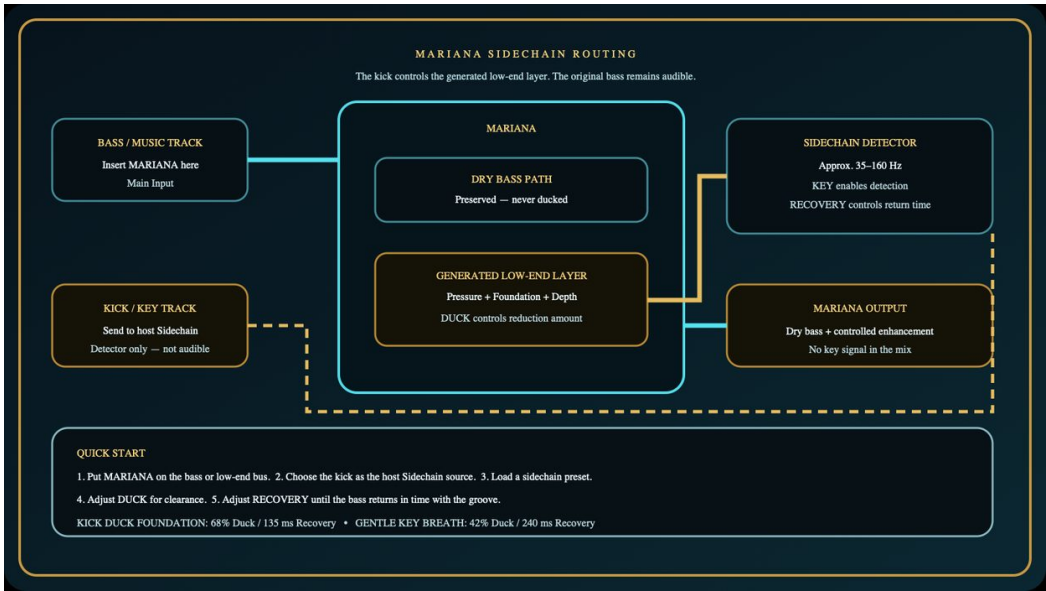
MARIANA exposes an optional stereo host Sidechain bus. Its 35-160 Hz detector uses a 5 ms attack and user-controlled 40-420 ms recovery; the external key is never mixed into the output.

Reduction affects only MARIANA's enhancement delta, preserving dry bass. KEY off is sample-path neutral. The original 14 programs keep KEY off; KICK DUCK FOUNDATION loads 68 percent / 135 ms and GENTLE KEY BREATH loads 42 percent / 240 ms.

Factory programs

MARIANA SWEET SPOT; LOW GRAVITY CLEAN; WARM SUBFIELD; SUB WEIGHT; DEEP PRESSURE; ABYSSAL FIELD; TRENCH FLOOR; GENTLE FINISH; OCEAN FLOOR; SUB ENHANCED; KICK PRESSURE; ABYSS BLOOM; TIDAL TRACE; SHALLOW CURRENT; KICK DUCK FOUNDATION; GENTLE KEY BREATH

Layer-safe sidechain routing



MONOLITH

Low-end impact and body shaper with excavation, weight, resonance, reverse motion, drive, balance, and an optional kick-key sidechain that clears only the added stone/weight layer.

Declared product: MONOLITH | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Low End Source | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
gate	gate	Referenced
input	input	Referenced
output	output	Referenced
channelmode	channelmode	Referenced
gainmatch	gainmatch	Referenced
excavation	excavation	Referenced
impact	impact	Referenced
weight	weight	Referenced
resonance	resonance	Referenced
reverse	reverse	Referenced
drive	drive	Referenced
balance	balance	Referenced
sidechain_enable	Sidechain Key	Bool
sidechain_duck	Stone Duck	Float
sidechain_recovery	Key Recovery	Float

Artwork authority

source/monolith_bg.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT / OUTPUT	Live pre/post level	Use for bypass and level matching.
IMPACT	Low-end transient activity	Shows the punch component being produced.
WEIGHT	Sustained low-frequency body	Shows foundation rather than peak attack.
RESONANCE	Active resonant low-end energy	Persistent high readings warn of ringing.
REVERSE	Reverse-motion contribution	Shows how strongly the reverse path is active.
Monolith field	Combined low-end activity map	Reactive summary, not a spectrum analyzer.
KEY	Detected 40-180 Hz sidechain energy	Moves only when KEY is enabled and a host sidechain is routed.
STONE DUCK	Reduction of the added stone/weight layer	The original input impact remains outside this reduction.

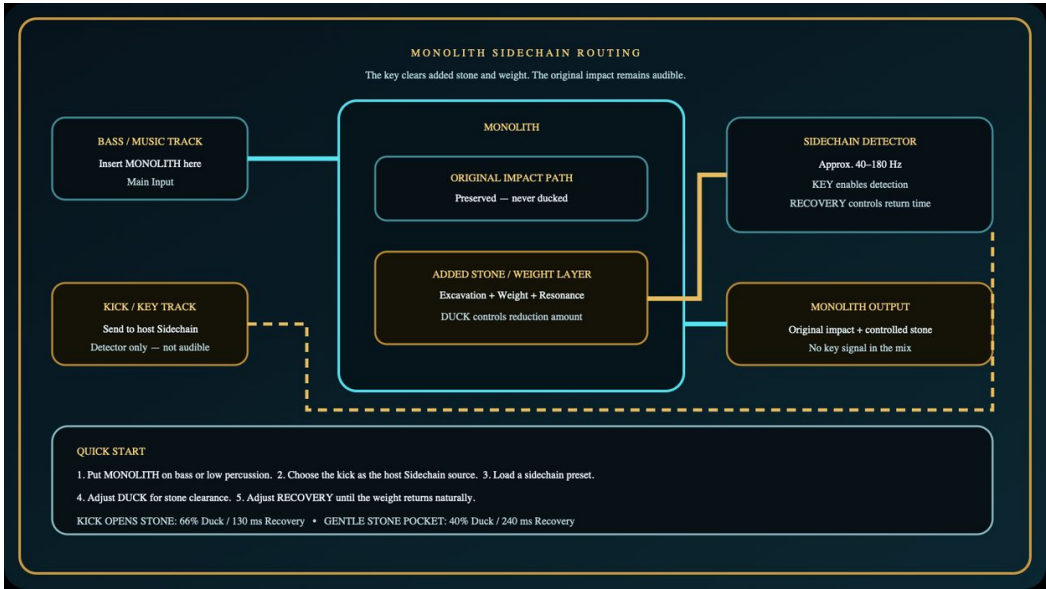
MONOLITH exposes an optional stereo host Sidechain bus. Its detector is band-limited to approximately 40-180 Hz with a 5 ms attack and user-controlled 40-420 ms recovery.

Reduction is applied to the exact wet-minus-dry stone/weight delta before recombination, preserving the original input impact. KEY off is sample-path neutral and adds no lookahead or reported latency.

Factory programs

MONOLITH PHI IMPACT; MONOLITH TRANSIENT STONE; MONOLITH BODY WEIGHT; MONOLITH SHAPE IMPACT; MONOLITH BUS CARVE; MONOLITH GATEWAY; KICK OPENS STONE [Stereo]; GENTLE STONE POCKET [Stereo]

Layer-safe sidechain routing



Depth Boost Pro

Focused low-end depth extension built on delay-based mid/side field placement. Primary role is low-end depth and width; secondarily it is a spatial gain stage, with a dedicated Spatial Gain control (0 dB down to -12 dB) applied after the depth field, plus protected input trim.

Declared product: DEPTH_BOOST_PRO | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Low End Source | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
input	Input Trim	Float
gainmatch	Gain Match	Bool
depth	Spatial Depth	Float
output	Spatial Gain	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

Assets/Images/Boost.png

Meter and visualizer semantics

Display	What it measures	How to read it
Touch sphere	Depth field plus live level	Water motion follows field/level activity; sphere position remains the output control.
Spatial field	Depth, field energy, and motion	A reactive placement view, not a calibrated spectrum.

NTRPL

Low-end transformation engine for rewrite map, novelty, interval motion, contour, shape, space, originality guard, and an optional kick-key sidechain that clears only the generated rewrite layer.

Declared product: NTRPL | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Low End Source | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
power	Power	Bool
gainmatch	Gain Match	Bool
chordMap	Rewrite Map	Choice
interpolate	Novelty	Float
motif	Chord Rewrite	Float
rhythm	Motion	Float
pitch	Interval Distance	Float
melody	Contour	Float
shape	Shape	Float
field	Space	Float
guard	Originality Guard	Float
alternateMix	Mix	Float
output	Output	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
sidechain_enable	Sidechain Key	Bool
sidechain_duck	Rewrite Duck	Float
sidechain_recovery	Key Recovery	Float

Artwork authority

assets/ntrpl.png

Meter and visualizer semantics

Display	What it measures	How to read it
SOURCE	Original input contribution	Shows how much source survives into the main route.
REWRITE	Generated harmonic-rewrite activity	Shows the newly synthesized low-end material.
ALT OUT	Alternative new-signal output	Always represents the generated signal path.
MAIN	Final blended output	Use against SOURCE for level and blend context.
Harmonic Rewriter	Live input-to-output contour	Shows source energy changing into rewritten energy.
Rewrite Map	Novelty, chord, interval, contour, motion, space, guard, and mix state	A live process map; it is not a frequency analyzer.

Display	What it measures	How to read it
KEY	Detected 35-160 Hz sidechain energy	Moves only when KEY is enabled and a host sidechain is routed.
REWRITE DUCK	Reduction of the generated rewrite layer	The source bass path remains outside this reduction.

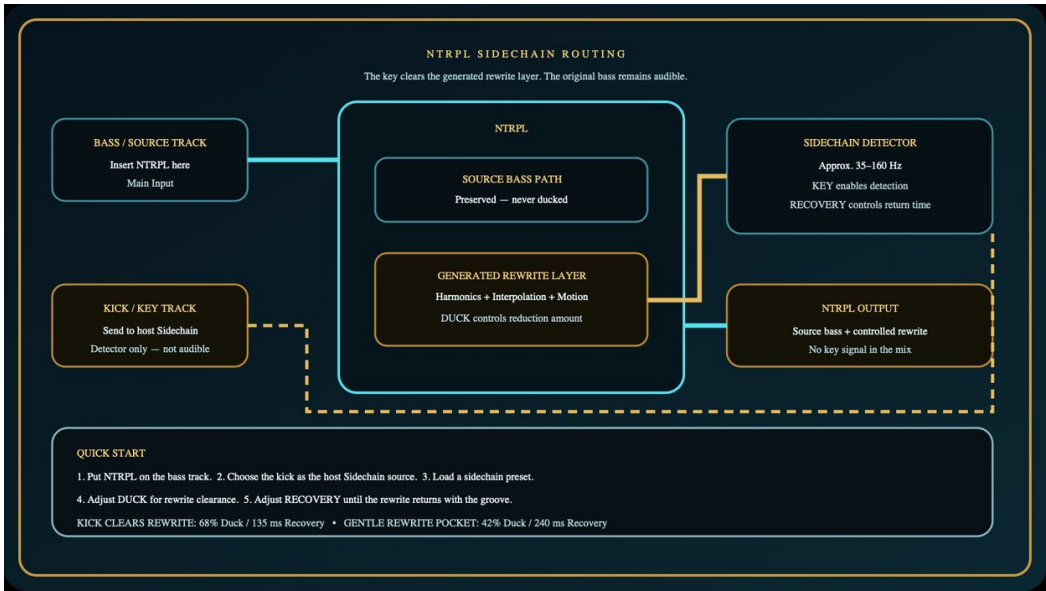
NTRPL exposes an optional stereo host Sidechain bus. Its key detector is band-limited to approximately 35-160 Hz with a 5 ms attack and user-controlled 40-420 ms recovery. The key is never mixed into the output.

Sidechain reduction multiplies only the generated rewrite/alternate layer. The source-retained main component stays outside the duck multiplier. KEY off is sample-path neutral and adds no lookahead or reported latency.

Factory programs

KICK CLEARS REWRITE [Stereo]; GENTLE REWRITE POCKET [Stereo]

Layer-safe sidechain routing



DRUM INSTRUMENT Technical Manual

Drum creation, sampling, sequencing, and impact placement

Create and shape drum material, then establish the musical impact and crowd-response field.

Included products: DIH Drum Morpher, BALLPARK, DrumKrushGlue

Routing authority: Drum route: DIH Drum Morpher -> Ballpark -> DrumKrushGlue.

Canon baseline

Normalized control 1 equals 0.809016994 (PHI/2). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

DIH Drum Morpher

Sixteen-pad drum sampler and host-synced sequencer with sample loading, pad shaping, restored factory programs, pattern memory, capture, morph, and performance controls. The current AU is registered and uses smoothed output normalization.

Declared product: Drum | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Instrument Track / Drum Source | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
internal	16 velocity-sensitive pads and MIDI notes	Custom runtime control
internal	WAV, AIFF, CAF, and MP3 sample loading	Custom runtime control
internal	Volume, pitch, morph, filter, punch, air, and ADSR per pad	Custom runtime control
internal	Host tempo, swing, grid rate, steps, play, and record	Custom runtime control
internal	Global morph, metamorphosis, momentum bend, governor, and ceiling	Custom runtime control
internal	Sampler capture, playback, and transfer to pad	Custom runtime control
internal	Sixteen per-pad mute toggles for isolating or removing kit voices	Custom runtime control

Artwork authority

source/DrMorph.png

Meter and visualizer semantics

Display	What it measures	How to read it
L / R output	Left and right output activity	Confirms stereo drum output and imbalance.
GOVERNOR	Output containment activity	More activity means protection is holding energy back.
MORPH	Current metamorphosis activity	Shows how strongly the active pads are transforming.
Pad waveform	Loaded sample amplitude over time	Use to identify attacks, tails, and silence in each pad.

OutputNormalizer is prepared before processing and advances rampStep per sample, preventing block-stepped gain changes. The installed Audio Unit passes registration/validation and exposes the restored factory program bank.

BALLPARK

Drum impact and placement processor whose base progression and hit calls turn performance energy into an audible stadium-response field.

Declared product: Ballpark | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Drum Track / Drum Bus | **Channel mode:** not detected | **Dual-mono link:** not detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
internal	Base mode and crowd-response progression	Custom runtime control
internal	Mound gain and switch-hitter transient mode	Custom runtime control
internal	Drive, depth, width, focus, clarity, mix, and convergence	Custom runtime control
internal	Input, RMS, peak, gain-reduction, and ceiling meters	Custom runtime control
internal	Foul, walk, single, double, triple, and home-run hit calls	Custom runtime control

Artwork authority

Assets/Images/Ballpark_Wide_Stadium.png

Meter and visualizer semantics

Display	What it measures	How to read it
IN	Input level	Confirms drum energy entering the stadium model.
GR	Gain reduction	Shows active impact containment.
OUT	Output level	Use with IN for level matching.
RMS / PEAK	Body level and fast peak	Their separation describes drum crest and punch.
Ceiling score / hit call	Composite impact, control, and ceiling score	Walk through Home Run describes response intensity, not sports prediction.

Factory programs

ON; GAN

DrumKrushGlue

Bus-preferred drum cohesion stage for binding hits into one controlled impact field, with an optional kick-key sidechain that opens a pocket only in its added weight and glue layer.

Declared product: DrumKrushGlue | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Drum Bus / Parallel Drum Impact Return | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
power	Power	Bool
gainmatch	Gain Match	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
sidechain_enable	Sidechain Key	Bool
sidechain_duck	Pocket Duck	Float
sidechain_recovery	Key Recovery	Float

Artwork authority

Assets/Images/DrumKrushGlue.png

Meter and visualizer semantics

Display	What it measures	How to read it
IMPACT	Detected drum peak activity	Shows hit strength entering the impact path.
BODY	Sustained drum energy	Shows weight carried between hits.
REPAIR	Glue/control activity	Shows how strongly cohesion is correcting the bus.
CEILING	Protected output pressure	Persistent maximum activity means the chain is overdriven.
KEY	Detected 40-180 Hz sidechain energy	Moves only when KEY is enabled and a host sidechain is routed.
POCKET	Reduction of the added weight/glue layer	The original drum path remains outside this reduction.

DrumKrushGlue exposes an optional stereo host Sidechain bus. Its detector is band-limited to approximately 40-180 Hz with a 4 ms attack and user-controlled 40-420 ms recovery.

Reduction is applied only to the processed-minus-input Krush/Weight/Glue delta. The main drum path remains outside the duck multiplier. KEY off is sample-path neutral and adds no lookahead or reported latency.

Factory programs

KICK POCKET GLUE [Stereo]; GENTLE BUS POCKET [Stereo]

COLOR AMP TAPE Technical Manual

Saturation, amplifier, tape, and harmonic color

Add character through distinct amp, tube, tape, and body-focused engines.

Included products: APIS, PtahTube, KHEPRI, SOVEREIGN

Routing authority: Music/color route: KHEPRI -> PTAHTUBE -> APIS. SOVEREIGN is a source-oriented amp alternative.

Canon baseline

Normalized control 1 equals 0.809016994 (PHI/2). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

APIS

Body and color processor built around color, body, petal dispersion, air, protected I/O, and channel-mode support.

Declared product: \${PROJECT_NAME} | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
INPUT	Input	Float
COLOR	Color	Float
BODY	Body	Float
FLOWER	Flower Petal	Float
AIR	Air	Float
OUTPUT	Output	Float
channelmode	Channel Mode	Float
dualmonolink	Dual Mono Link	Float
gainmatch	Gain Match	Bool

Artwork authority

assets/background_stripped.png

PtahTube

Phi-based tube color processor with input, drive, energy, balance, focus, output, and Flower of Life dispersal tied to its illuminated tube field.

Declared product: PTAHTUBE | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
input	Input	Float
drive	Drive	Float
output	Output	Float
energy	Energy	Float
balance	Balance	Float
power	Power	Float
focus	Focus	Float
flower	Flower of Life Dispersal	Bool
gainmatch	Gain Match	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

Assets/Ptah.png

Factory programs

MANUAL | USER SETTINGS

KHEPRI

Sacred tape machine with selectable machine, Phi formula, tape type (Gold Alloy, Ferric Oxide, Metal Alloy), speed, drive, bias, wow, flutter, head bump, HF rolloff, and parallel mix.

Declared product: Khepri | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
power	Power	Bool
gainmatch	Gain Match	Bool
machine	Machine	Choice
formula	Phi Formula	Choice
tapeType	Tape Type	Choice
speed	Speed	Choice
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
input	Input	Float
drive	Drive	Float
bias	Bias	Float
wow	Wow	Float
flutter	Flutter	Float
bump	Head Bump	Float
rolloff	HF Rolloff	Float
mix	Mix	Float
output	Output	Float

Artwork authority

Assets/Khepri_Background.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT VU	Ballistic input level	Calibrated to the printed analog VU scale.
OUTPUT VU	Ballistic post-tape level	Compare with INPUT to avoid loudness bias.
Waveform	Recent live output samples	Shows real signal motion, not a decorative animation.
Reels / beetle glow	Tape motion and drive state	Operational feedback; not a calibrated level meter.

Metal Alloy's high-frequency ceiling is constrained to 45 percent of the active sample rate before coefficient generation, keeping the filter below Nyquist at every supported host rate. A final finite-sample containment guard resets local tape memory and suppresses an invalid sample if upstream state ever becomes non-finite.

Factory coverage now spans all three tape stocks: Gold Alloy for balanced reference and glue, Ferric Oxide for vocal/body/vintage uses, and Metal Alloy for clean print, drum transient, and air uses. The former Full Reel and Low Reel extremes are not part of the current factory program table.

SOVEREIGN

Source-oriented amplifier and tone stage with drive, bass, mid, treble, presence, and master controls.

Declared product: \${PROJECT_NAME} | **Version:** 3.15 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Mix Bus | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
power	Power	Bool
drive	Drive	Float
bass	Bass	Float
mid	Mid	Float
treble	Treble	Float
presence	Presence	Float
master	Master	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool

Artwork authority

source/Sovereign.png

FIELD TONE BALANCE Technical Manual

Field tone, restoration, and tonal balance

Rebalance or restore a source before it reaches later spatial or mastering stages.

Included products: AUSET, AUSAR

Routing authority: Auset is bus preferred for field tilt. Ausar is bus preferred for restoration work.

Canon baseline

Normalized control 1 equals 0.809016994 (PHI/2). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

AUSET

Field-tone balancer with intensity, tilt, power, and optional Flower of Life dispersal.

Declared product: AUSET | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Bus Preferred | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
INTENSITY	Intensity	Float
TILT	Tilt	Float
POWER	Power	Bool
gainmatch	Gain Match	Bool
FLOWER_DISPERSAL	Flower of Life Dispersal	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

assets/auset_bg.png

Factory programs

AUSET FULL RANGE; AUSET NIGHT BALANCE; AUSET LUNAR LIFT; AUSET SOLAR REVEAL; AUSET VELVET BODY; AUSET SHADOW DEPTH; AUSET AIR LIFT REVEAL; AUSET MASTER SUPPORT; AUSET FLOWER BLOOM

AUSAR

Restoration processor for hum repair, broadband extraction, transient integrity, and controlled output.

Declared product: AUSAR | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Bus Preferred | **Channel mode:** not detected | **Dual-mono link:** not detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
gainmatch	Gain Match	Bool
HUM_REPAIR	Hum Repair	Float
BROADBAND_EXT	Broadband Extraction	Float
TRANSIENT_INT	Transient Integrity	Float
OUTPUT_GAIN	Output	Float

Artwork authority

assets/ausar_bg.png

Factory programs

AUSAR SOURCE RESTORE; AUSAR RELIC WASH; AUSAR RESURRECTION FIELD; AUSAR HUM DEEP CLEAN; AUSAR BROADBAND REVEAL; AUSAR TRANSIENT REPAIR; AUSAR ARCHIVAL BALANCE; AUSAR FULL RANGE PROOF

DIMENSIONAL IMPACT MOVE Technical Manual

Dimension, width, impact, and motion

Shape depth, front-to-back position, movement, and spatial delay as a coordinated field.

Included products: Depth Impact Hybrid X, PARALLAX, NSR

Routing authority: Width/depth route: Depth Impact Hybrid X -> Parallax -> NSR.

Canon baseline

Normalized control 1 equals 0.809016994 ($\text{PHI}/2$). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

Depth Impact Hybrid X

Six-mode spatial field for clarity, depth, low bloom, front push, hybrid pressure, fold, freeze, and source-to-horizon movement.

Declared product: Depth | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
output	Output	Float
mode	Hybrid X Mode	Choice
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
horizonFold	Horizon Fold	Bool
freeze	Event Freeze	Bool
echo	Echo	Bool
cpuSaver	CPU Saver	Bool
bypass	Bypass	Bool
gainmatch	Gain Match	Bool
spectralClarity	spectralClarity	Referenced
spatialDepth	spatialDepth	Referenced
loContour	loContour	Referenced
subPressure	subPressure	Referenced
foundation	foundation	Referenced
ground	ground	Referenced
mass	mass	Referenced
presenceLift	presenceLift	Referenced
frontPush	frontPush	Referenced
dimensionalGain	dimensionalGain	Referenced
extension	extension	Referenced
horizonDrive	horizonDrive	Referenced
horizonMaximizer	horizonMaximizer	Referenced
eventSmear	eventSmear	Referenced
eventGravity	eventGravity	Referenced
hybridAmount	hybridAmount	Referenced
immersionBlend	immersionBlend	Referenced
marianaBlend	marianaBlend	Referenced
boostBlend	boostBlend	Referenced
horizonBlend	horizonBlend	Referenced
eventBlend	eventBlend	Referenced
pressureBlend	pressureBlend	Referenced

Artwork authority

Meter and visualizer semantics

Display	What it measures	How to read it
RMS FIELD	Overall sustained field energy	Shows the body of the dimensional result.
LEFT / RIGHT	Per-channel output activity	Use to spot unintended channel imbalance.
HORIZON	Depth and distant-field contribution	Higher values indicate stronger rearward space.
EVENT	Transient/front-push contribution	Higher values indicate stronger forward impact.
Immersion / Foundation / Motion	Three-part dimensional activity summary	Read together to distinguish depth, low support, and movement.

PARALLAX

Bus-preferred width and depth field with focus, spread, axis core, link, mix, and output control.

Declared product: PARALLAX | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Bus Preferred | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
width	Width	Float
depth	Depth	Float
focus	Focus	Float
spread	Stereo Spread	Float
axis	Axis Core	Float
mix	Mix	Float
output	Output	Float
field_link	Field Link	Bool
power	Power	Bool
gainmatch	Gain Match	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

assets/parallax_preset_00_deep_space.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT / OUTPUT	Live pre/post level	Use for level matching around the width field.
Field visualizer	Width, depth, focus, spread, and axis state	Motion shows spatial behavior; it is not a frequency plot.

Factory programs

Deep Space; Wormhole; Nebula Shift; Stellar Grid; Planetary Orbits; Light Speed; Quantum Field; Dimensional Gate; Gravity Well; Ascension

NSR

Phi-ratio spatial delay and movement field combining travel, microphase, air, anchor, spread, time delay, flange, and mix.

Declared product: NSR | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Bus Preferred | **Channel mode:** not detected | **Dual-mono link:** not detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
gainmatch	Gain Match	Bool
travel	travel	Referenced
microphase	microphase	Referenced
air	air	Referenced
anchor	anchor	Referenced
spread	spread	Referenced
field	field	Referenced
timeDelay	timeDelay	Referenced
flange	flange	Referenced
mix	mix	Referenced

Artwork authority

Resources/NSR_bg.png

Meter and visualizer semantics

Display	What it measures	How to read it
SPECTRUM	Energy across displayed frequency bands	Shows tonal distribution of the current signal.
CORRELATION	Left/right phase similarity	+1 is mono-coherent, 0 is decorrelated, negative values risk cancellation.
L / R	Channel activity	Use to detect imbalance before and after the field.
PHI FIELD VIEW	Combined spatial-field intensity	Shows overall activity of the responder.
STEREO FIELD / WIDTH	Side-to-mid spatial width	Check correlation whenever width stays high.
FIELD ACTIVITY	Wet spatial-path energy	Shows how much processed field is actually active.
DRY / WET waveform	Recent source and processed traces	Compare timing, shape, and motion between paths.
Signal Path	Activity at each processing stage	Locates where energy enters, moves, feeds back, and is contained.

FX SPATIAL Technical Manual

Delay, reverb, tail, and spatial environment

Create audible space, movement, and controlled tails on parallel spatial lanes.

Included products: NTRSTLLR, Cathedral, SacredVerb

Routing authority: Parallel spatial route: NTRSTLLR -> Cathedral -> SacredVerb.

Canon baseline

Normalized control 1 equals 0.809016994 (PHI/2). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

NTRSTLLR

Audible interstellar delay and tail engine with travel, warp, orbit, portal, delay, air, haze, width, diffusion, axis, and Nebula/Wormhole modes.

Declared product: NTRSTLLR | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Bus Preferred | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call not detected; integration review required before release claims
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
trumpet_l	Micro Peak	Float
trumpet_c	Meso Density	Float
trumpet_r	Macro Level	Float
makeup_gain	Makeup Gain	Float
travel	Travel	Float
warp	Warp	Float
orbit	Orbit	Float
portal	Portal	Float
time_delay	Time/Delay	Float
rubber	Rubber	Float
air	Air	Float
haze	Haze	Float
width	Width	Float
diffusion	Diffusion	Float
axis	Axis	Float
mix	Mix	Float
output	Output	Float
wormhole_mode	Wormhole Mode	Bool
nebula_mode	Nebula Mode	Bool
phi_grid	Phi Grid	Bool
power	Power	Bool
gainmatch	Gain Match	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

assets/NTRSTLLR_BG_20260715.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT / OUTPUT	Weighted peak-plus-RMS level	Use for level matching around the spatial return.

Display	What it measures	How to read it
TONE RESPONSE	Current low-to-high response shape	Shows tonal contour, not a mastering EQ recommendation.
WORMHOLE	Delay/warp-path energy	Shows active wormhole return strength.
NEBULA	Portal/diffusion-path energy	Shows active nebula tail strength.
SINGULARITY	Axis and pull concentration	Higher values indicate stronger inward focus.
PHI FIELD OBSERVATION	Combined macro-control activity	A spatial-state summary rather than loudness.
PHI ECHO / DELAY FIELD	Delay taps, orbit, and return motion	Use to understand tail movement and density.

Cathedral

Seven-river chamber environment with arch, convergence, 3-6-9 nodes, protected I/O, and parallel mix.

Declared product: Cathedral | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Bus Preferred | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
arch	Arch	Float
rivers	Seven Rivers	Float
nodes	3-6-9 Nodes	Float
convergence	Convergence	Float
input	Input	Float
mix	Mix	Float
output	Output	Float
channel_mode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool

Artwork authority

Assets/Images/Cathedral.png

Meter and visualizer semantics

Display	What it measures	How to read it
ENERGY	Current chamber RMS in dB	Shows how strongly the room is being excited.
DIFFUSION	3-6-9 node density	Higher values produce a denser, smoother tail.
WIDTH	Node density multiplied by wet mix	Shows audible spatial spread, not dry-source width.
RT60	Estimated mid-band decay time	Seconds required for the modeled room to decay by 60 dB.
ROOM RESONANCE	Modeled low/mid/high response	Use to identify room emphasis before printing.

SacredVerb

Sacred spatial reverb with pre-delay, chamber size, damping, rate, depth, Neter blend, resonance, chamber and pattern modes, and protected presets.

Declared product: SacredVerb | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Bus Preferred | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call not detected; integration review required before release claims
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
PRE_DELAY	PRE DELAY	Float
ROOM_SIZE	CHAMBER SIZE	Float
DAMPING	DAMPING	Float
REVERB_MIX	WET MIX	Float
MOD_RATE	RATE	Float
MOD_DEPTH	DEPTH	Float
NETER_BLEND	NETER BLEND	Float
RESONANCE	RESONANCE	Float
INPUT	INPUT	Float
OUTPUT	OUTPUT	Float
CHAMBER	Chamber	Int
PATTERN_MODE	Pattern Mode	Int
FLOWER	Flower of Life	Bool
ECO_MODE	Eco Mode	Bool
BYPASS	Bypass	Bool
gainmatch	Gain Match	Bool
TRIG_TAIL	Trigger Tail	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
TRIG_	TRIG_	Referenced
EN_	EN_	Referenced

Artwork authority

Assets/Images/SacredVerb.png

Meter and visualizer semantics

Display	What it measures	How to read it
Seven Hathor meters	Energy at enabled Solfeggio nodes	Each column shows its resonant band activity and peak hold.
Output meter	Current reverb-output peak	Use to keep the return controlled and level matched.
3-6-9 modulator	Active modulation/pattern state	Shows motion settings; it is not a loudness meter.

HARMONICS PITCH Technical Manual

Harmonic structure, pitch, and instrument generation

Shape harmonic emphasis, rhythmic spectral motion, or generated instrument voices.

Included products: THOTH, TEHUTI, AU

Routing authority: Thoth and Tehuti are insert safe. AU is the family instrument and synth.

Canon baseline

Normalized control 1 equals 0.809016994 ($\text{PHI}/2$). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

THOTH

Sacred harmonic EQ with low and high shelves, visible shelf curves, Flower of Life mode, protected I/O, and parallel mix.

Declared product: Thoth | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
power	Power	Bool
input	Input	Float
mix	Mix	Float
ls_gain	Low Shelf	Float
hs_gain	High Shelf	Float
output	Output	Float
flower	Flower of Life	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool

Artwork authority

source/Thoth.png

Meter and visualizer semantics

Display	What it measures	How to read it
Spectrum bands	Live energy in each displayed EQ region	Use to see where harmonic energy is accumulating.
Shelf curves	Current low/high shelf response	These show the selected EQ shape, not measured level.
IN / OUT	Pre/post peak activity	Use for level-fair harmonic decisions.

TEHUTI

Harmonic motion processor using low, mid, and high ducking with depth, speed, mix, and Flower of Life mode.

Declared product: \${PROJECT_NAME} | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
power	Power	Bool
low_duck	Low Duck	Float
mid_duck	Mid Duck	Float
high_duck	High Duck	Float
depth	Depth	Float
speed	Speed	Float
mix	Mix	Float
flower	Flower of Life	Bool
gainmatch	Gain Match	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

source/Tehuti.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT	Main-program peak activity	Shows the signal being ducked.
SIDECHAIN SEND	External sidechain envelope	Confirms that the host sidechain is present.
LOW GR	Reduction below 162 Hz	Shows low-band ducking driven by the sidechain.
MID GR	Reduction from 162 Hz to 1.62 kHz	Shows mid-band ducking driven by the sidechain.
HIGH GR	Reduction above 1.62 kHz	Shows high-band ducking driven by the sidechain.
Duck curves	Expected reduction versus sidechain energy	The live dot shows the current operating point.
History	Recent sidechain and reduction motion	Use to judge pumping and release behavior.

AU

DIHWorld instrument and synth with factory sound selection, drag-and-drop source support, nine Neter voices, modulation, space, glide, sustain, and master control. The current AU is registered and uses smoothed output normalization.

Declared product: AU | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Instrument Track | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
SOUND	Factory Sound	Choice
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
PTAH	PTAH	Referenced
RA	RA	Referenced
SET	SET	Referenced
ASAR	ASAR	Referenced
MAAT	MAAT	Referenced
SEKHMET	SEKHMET	Referenced
KHNUM	KHNUM	Referenced
HERU	HERU	Referenced
DRIVE	DRIVE	Referenced
AMOUNT	AMOUNT	Referenced
DEPTH	DEPTH	Referenced
SV_MIX	SV_MIX	Referenced
HARP	HARP	Referenced
ORGAN	ORGAN	Referenced
CHOIR	CHOIR	Referenced
MOD	MOD	Referenced
BREATH	BREATH	Referenced
VIBRATO	VIBRATO	Referenced
PHI	PHI	Referenced
AZIMUTH	AZIMUTH	Referenced
FLOWER	FLOWER	Referenced
HATHOR	HATHOR	Referenced
RAW	RAW	Referenced
MOD_MACRO	MOD_MACRO	Referenced
MASTER	MASTER	Referenced
GLIDE	GLIDE	Referenced
PURE	PURE	Referenced
SUSTAIN	SUSTAIN	Referenced
GAINMATCH	GAINMATCH	Referenced

Artwork authority

Assets/AU_Background.png

OutputNormalizer is prepared before processing and advances rampStep per sample, preventing block-stepped gain changes. The installed Audio Unit passes registration/validation.

CREATIVE TRANSFORM Technical Manual

Creative reconstruction and transformation

Recombine processors, recover hidden content, or reshape a source beyond corrective work.

Included products: SlotMachine, Archaeologist X, SERQET

Routing authority: Use as deliberate creative stages; audition in context and bypass match before printing.

Canon baseline

Normalized control 1 equals 0.809016994 (PHI/2). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

SlotMachine

Three-chamber creative host that combines source, opposition, and resolution through architecture, blend, morph, and containment.

Declared product: SlotMachine | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Creative Transform | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
chamberA	Chamber A Source	Choice
chamberB	Chamber B Opposition	Choice
chamberC	Chamber C Resolution	Choice
architecture	Architecture	Float
blend	Blend	Float
morph	Morph	Float
containment	Containment	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool

Artwork authority

assets/Images/SlotMachine_BG.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT	Input peak	Confirms level entering the three chambers.
OUTPUT	Output peak	Use for level matching.
RMS	Average program body	Shows sustained density.
CREST	Peak-to-RMS relationship	Higher values mean sharper transients; lower values mean denser material.

Archaeologist X

Forensic recovery processor for excavation, recovery field, harmonic extraction, layer selection, gate, and parallel mix.

Declared product: ARCHAEOLOGIST | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Creative Transform | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
power	Power	Bool
gainmatch	Gain Match	Bool
excavation	Excavation Engine	Float
recovery	Recovery Field	Float
harmonic	Harmonic Extraction	Float
forensic	Forensic Layer	Float
gate	Gate	Float
mix	Mix	Float
output	Output	Float
profile	Profile	Choice
mode	Mode	Choice
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

Assets/archaeologist_x_bg_v2.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT / OUTPUT	Live pre/post peak	Use for bypass and level matching.
EXCAVATION	Low/deep recovery activity	Shows energy being uncovered by excavation.
RECOVERY	Recovered-source contribution	Shows how much repaired material is active.
HARMONIC	Extracted harmonic activity	Shows reconstructed tonal information.
FORENSIC	Forensic layer/gate activity	Shows the selective analysis path at work.
Observation Field	Combined four-process activity	A process summary, not a spectrum.

SERQET

Creative sting and width processor with intensity, venom saturation, convergence focus, and output control.

Declared product: SERQET | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Creative Transform | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
STING	Width Reach	Float
VENOM	Movement Trace	Float
FOCUS	Prism Focus	Float
OUTPUT	Output Master	Float
gainmatch	Gain Match	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

assets/SERQET_0006.png

Meter and visualizer semantics

Display	What it measures	How to read it
RMS WIDTH	Input RMS feeding the width field	Shows source energy available for widening.
PRISM WIDTH	Measured side-field width	Higher values indicate a broader stereo prism.
OUTPUT FIELD	Post-process RMS energy	Use with RMS WIDTH for level context.
Prism visualizer	Width, motion, focus, and output state	The polygon is a spatial-state map, not a loudness target.

Factory programs

SERQET CANON WIDTH; SERQET VEIL MOTION; SERQET PRISM LENS; SERQET WIDE FIELD; SERQET BUS VEIL

MASTERING CORE Technical Manual

Mastering preparation, balance, lift, authority, and containment

Prepare, balance, polish, lift when needed, and contain the final program.

Included products: ANKH, SPHYNX, MAAT, Horizon X, PLATINUM

Routing authority: Choose one preparation stage, one balance or field stage, then ANKH. Place SESHAT last as observation only.

Canon baseline

Normalized control 1 equals 0.809016994 (PHI/2). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

Shell canon: Mastering Core uses a silver preset-shell stripe and machined silver/chrome frame. PLATINUM is a primary Mastering Core product with Color / Amp / Tape support as a secondary role.

ANKH

Sacred optical leveler for late-stage compression, Ma'at limiting, attack, release, drive, ratio, stereo link, Flower luminosity, and final containment.

Declared product: ANKH | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Master Field | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
gainmatch	Gain Match	Bool
inputDb	RA Input	Float
mix	Mix	Float
attack	Sekhem Attack	Float
release	Ma'at Release	Float
drive	Ptah Drive	Float
ascension	Ascension Ratio	Float
outputDb	Amun Output	Float
compress	Compress	Bool
limit_on	Limit	Bool
flower_of_life	Flower of Life	Bool
link	Stereo Link	Bool
bypass	Bypass	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

assets/ankh_bg.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT LEVEL	Peak level entering ANKH	Use to verify healthy mastering-chain entry.
GAIN REDUCTION	Current optical/limiter attenuation	More fill means more level is being removed.
OUTPUT LEVEL	Peak level after containment	Compare with Input for level-fair decisions.

SPHYNX

Mastering normalizer and calibrator that prepares input level, target, engine strength, normalization amount, mix, and output.

Declared product: SPHYNX | Version: 3.22 | Customer formats: macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Mix Bus | Channel mode: detected | Dual-mono link: detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
INPUT	Input	Float
CALIBRATE	Calibrate	Float
NORMALIZE	Normalize	Float
ENGINE	Engine	Float
TARGET	Target	Float
MIX	Mix	Float
OUTPUT	Output	Float
gainmatch	Gain Match	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

assets/sphynx.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT PEAK	Peak entering calibration	Confirms available headroom before normalization.
PROGRAM RMS	Average program energy	Use to understand body level, not final delivery loudness alone.
AUTO TRIM	Gain correction calculated by SPHYNX	Center is 0 dB; left cuts and right boosts.
OUTPUT PEAK	Peak after normalization	Verify headroom after calibration.
Crest / headroom	Peak-to-RMS and remaining space	Use together to avoid flattening transient structure.

MAAT

Mastering balance processor for foundation and air, bandwidth, truth, architecture, phase, stereo link, saturation mode, and output.

Declared product: Maat | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Mix Bus | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call not detected; integration review required before release claims
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
FOUNDATION_BOOST	Foundation Boost	Float
FOUNDATION_ATTEN	Foundation Atten	Float
FOUNDATION_FREQ	Foundation Frequency	Float
AIR_BOOST	Air Boost	Float
AIR_ATTEN	Air Atten	Float
BANDWIDTH	Bandwidth	Float
BALANCE	Balance	Float
TRUTH	Truth	Float
ARCHITECTURE	Architecture	Float
OUTPUT	Output	Float
POWER	Power	Bool
gainmatch	Gain Match	Bool
PHASE_INVERT	Phase Invert	Bool
FOUNDATION_WEIGHT_ON	Foundation Weight On	Bool
RUMBLE_HPF	Rumble HPF	Float
STEREO_LINK_MODE	Stereo Link Mode	Float
SATURATION	Saturation Tube Tape	Float

Artwork authority

Assets/Images/Maat.png

Meter and visualizer semantics

Display	What it measures	How to read it
TRUTH	Composite tonal/phase balance score	Higher values mean the current foundation/air balance is closer to the intended field.
INPUT / OUTPUT	Live pre/post level	Use for level matching.
FOUNDATION	Low-frequency contribution	Shows active foundation energy.
AIR	Upper-frequency contribution	Shows active air energy.

Horizon X

Mastering finalizer with dawn/dusk states, volume, drive, maximizer, depth, clarity, gravity, pull, collapse, return, and blended Horizon/Event modes.

Declared product: Horizon | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Mix Bus | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
input	Input	Float
output	Output	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
bypass	Bypass	Bool
clip	RA Clip	Bool
gainmatch	Gain Match	Bool
volume	Volume	Factory
drive	Drive	Factory
maximizer	Maximizer	Factory
depth	Depth	Factory
clarity	Clarity	Factory
mix	Mix	Factory
gravity	Gravity	Factory
pull	Pull	Factory
collapse	Collapse	Factory
return	Return	Factory
eventBlend	Event Blend	Factory
horizonBlend	Horizon Blend	Factory
field	Field Of Life	Factory
mode	Mode	Factory

Artwork authority

assets/horizon_dawn.png

Meter and visualizer semantics

Display	What it measures	How to read it
LIVE SIGNAL	Per-channel input RMS on a -60 to 0 dBFS scale	Shows program energy entering the finalizer.
FIELD STATE	Combined Horizon/Event processing state	A process-state visualization, not a delivery-loudness meter.
OUTPUT LEVEL	Post-finalizer level on a -60 to 0 dBFS scale	Use with LIVE SIGNAL for level-fair review.

PLATINUM

Mastering authority processor using silver air, gold drive, copper body, iron glue, nickel focus, platinum authority, and final output.

Declared product: PLATINUM | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Mix Bus | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool
AG_AIR	AG Air	Factory
AU_DRIVE	AU Drive	Factory
CU_BODY	CU Body	Factory
FE_GLUE	FE Glue	Factory
NI_FOCUS	NI Focus	Factory
PT_AUTHORITY	PT Authority	Factory
OUTPUT	Output	Factory

Artwork authority

assets/platinum_bg.png

MASTER FIELD Technical Manual

Macro master-field structure and finish

Organize foundation, architecture, dimension, tonal axis, and finish at the macro level.

Included products: IMHOTEP

Routing authority: Imhotep belongs late in the clean mastering route before ANKH containment.

Canon baseline

Normalized control 1 equals 0.809016994 (PHI/2). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

IMHOTEP

Macro master-field builder for foundation, architecture, dimension, finish, tonal axis, I/O trim, and Flower of Life dispersal.

Declared product: IMHOTEP | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Master Field | **Channel mode:** detected | **Dual-mono link:** not detected

Source-review status

- Exact DIHEngine::processSample call not detected; integration review required before release claims
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
dispersal	Flower of Life Dispersal	Bool
gainmatch	Gain Match	Bool
channelmode	Channel Mode	Choice
foundation	Foundation	Factory
architecture	Architecture	Factory
dimension	Dimension	Factory
finish	Finish	Factory
tilt	Tonal Axis	Factory
input	Input Trim	Factory
output	Output Trim	Factory

Artwork authority

Assets/Images/Imhotep.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT PEAK	Peak entering the master builder	Confirms headroom before macro shaping.
FIELD ACTIVITY	Combined foundation, architecture, dimension, and finish activity	Shows how strongly the macro field is engaged.
OUTPUT / PHI CEILING	Output level relative to containment	Persistent maximum readings mean less available headroom.

Factory programs

MASTER BUILDER CANON; CLEAN PRINT FRAME; FOUNDATION MONUMENT; VOCAL MASTER ORDER; DENSE MIX REBUILD; OPEN PYRAMID FIELD; TRANSIENT STONEWORK; GREAT PYRAMID

UTILITY OBSERVATION Technical Manual

Observation, metering, and print confirmation

Observe final behavior without becoming a tone-changing stage.

Included products: SESHAT

Routing authority: Place Seshat at the end of the analysis and print lane.

Canon baseline

Normalized control 1 equals 0.809016994 (PHI/2). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

SESHAT

Final observation and analysis lane with restored spacious panel layout and optional peak hold; Full Observation shows the complete meter set, while each named preset shows only the meter or panel relevant to that view. It is not a tone-shaping processor.

Declared product: SESHAT | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Observation | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
hold_peaks	Hold Peaks	Bool
gainmatch	Gain Match	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

source/Seshat.png

Meter and visualizer semantics

Display	What it measures	How to read it
LUFS / BODY READ	Short-window perceptual loudness	Use for loudness context, not as the only delivery decision.
LEFT INPUT / RIGHT OUTPUT	Peak, RMS body, impact, and held peak	Shows entry and exit behavior; Hold Peaks retains the highest marker.
CORRELATION / MONO SAFETY	Left/right phase relationship	+1 is coherent, 0 is wide/uncorrelated, negative values risk cancellation.
SOLFEGGIO FREQUENCIES	Energy near the seven displayed resonant bands	Shows tonal concentration around 396 through 963 Hz.
IMPACT	Fast peak relative to the canon ceiling	Higher means stronger transient force.
FIELD	Body RMS plus stereo-side pressure	Higher means denser sustained spatial load.
MOMENTUM	Change in impact plus transient shape	Higher means faster, more active movement.
PHI ALIGNMENT	Proximity to DIH golden-ratio checkpoints	Higher means field pressure sits nearer a PHI reference point.
INTEGRITY	Similarity between impact and field pressure	Higher means transients and body remain proportionate.
CROWN	Weighted combined observation score	Summarizes impact, field, momentum, alignment, and integrity.
MOON	Transient shape plus momentum	Higher means stronger forward or launching behavior.
HF RISK	High-frequency energy imbalance	Higher values warn of potential harshness.
PHASE RISK	Correlation and side-energy danger	Higher values warn of mono cancellation.
LEVEL RISK	Peak/ceiling and transient danger	Higher values warn of insufficient headroom.

Display	What it measures	How to read it
COMFORT	Alignment, integrity, and low-risk balance	Higher values indicate a more stable field.
ADVISOR	Combined risk recommendation	A diagnostic suggestion; it does not change audio.
REEL 1 - Impact	Transient-force rank	First reel; 396 is low and MOON is highest.
REEL 2 - Field	Sustained spatial-pressure rank	Second reel; numbers are symbolic stages, not Hz.
REEL 3 - Momentum	Motion/transient-change rank	Third reel; higher stages mean more movement.
REEL 4 - Alignment	PHI-alignment rank	Fourth reel; higher stages mean closer alignment.
REEL 5 - Moon	Forward-propulsion rank	Fifth reel; higher stages mean stronger launch.

Preset view state controls panel visibility, not merely highlight color: Full Observation enables the full panel set; named views enable only their mapped meter/panel.